$$AB \neq BA.$$

I mention this because you will sometimes see the trace of the density matrix written as *Tr* **L***ρ*, instead of *Tr ρ***L**. These two expressions are equivalent.

Exercise 7.5:

a) Show that

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}^2 = \begin{pmatrix} a^2 & 0 \\ 0 & b^2 \end{pmatrix}.$$

b) Now, suppose

$$\rho = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{2}{3} \end{pmatrix}.$$

Calculate

$$\rho^2$$

$$Tr(\rho)$$

$$Tr(\rho^2).$$

c) If *ρ* is a density matrix, does it represent a pure state or a mixed state?

Exercise 7.6: Use Eq. 7.22 to show that if *ρ* is a density matrix, then